Ollscoil Mhá Nuad
Maynooth University

SEMESTER 1
2025-2026

CS401
Machine Learning & Neural Networks

Dr. T. De Wille, Dr. A. Mooney, Dr. C. Ennis

Time allowed: 2 hours

Answer at least ***three*** questions. For each question you must answer ***all sub-questions***.

Your mark will be based on your best ***three*** answers.

Where calculations are required, ***include your workings*** in your answer sheet.

All questions carry equal marks.

Instructions

	Yes	No	N/A
Formulae and Tables book allowed *(i.e. available on request)*		X	
Formulae and Tables book required *(i.e. distributed prior to exam commencing)*		X	
Statistics Tables and Formulae allowed *(i.e. available on request)*		X	
Statistics Tables and Formulae required *(i.e. distributed prior to exam commencing)*		X	
Dictionary allowed *(supplied by the student)*		X	
Non-programmable calculator allowed	X		
Students required to write in and return the exam question paper		X	

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1 ML Fundamentals + Evaluation [25 marks]

(a) Explain the fundamental difference between Supervised Learning, Unsupervised Learning, and Reinforcement Learning in terms of their training data and primary objective. For each, provide one classic example of a problem best suited for that paradigm. [5 marks]

(b) A hospital uses an algorithm to predict a very rare, high-risk disease (prevalence is 1 in 10,000 patients). The algorithm has a very high sensitivity (Recall = 99%) but a low specificity (False Positive Rate = 1%). [6 marks]

i) Why are metrics like Precision and Recall more informative than simple Accuracy in this extreme class imbalance scenario? *[3 marks]*

ii) Define and provide the formula for Precision. Explain why the hospital's patients who receive a positive test result will likely have a very low Precision. (No calculation required, just a conceptual explanation.) *[3 marks]*

(c) A data analyst is training a classification model to predict customer churn, experimenting with two models: [9 marks]

- Model X (Simple Logistic Regression): Achieves accuracy of 55% on the training set and 55% on the test set.
- Model Y (Decision Tree): Achieves accuracy of 99% on the training set and 60% on the test set.

i) Which of the two fundamental problems (high bias or high variance) is each model most likely suffering from? Justify your choice by referencing the performance metrics of the training and test sets. *[4 marks]*

ii) For Model Y, propose two distinct techniques that could be applied to mitigate its issue (from part i) and improve its performance on the test set. *[5 marks]*

(d) Briefly explain the role of each of the two main terms in the below cost function and their intended purpose in the context of preventing overfitting: [5 marks]

$$J(\theta) = \frac{1}{2m}\sum_{i=1}^{m}\left(\mathrm{h}_{(\theta)}\left(x^{(i)}\right) - y^{(i)}\right)^2 + \frac{\lambda}{2m}\sum_{j=1}^{n}\theta_j^2$$

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2 Reinforcement Learning and Fairness in ML [25 marks]

(a) Describe the MDP framework for modelling Reinforcement Learning problems and define the four core mathematical elements it comprises. Use the example of an RL autonomous driving agent to illustrate each element. [10 marks]

(b) What are the challenges, implications and trade-offs of a stochastic environment in the context of RL? [6 marks]

(c) What is meant by Bias and Fairness in the context of Machine Learning? [3 marks]

(d) Describe an example of best practice when it comes to mitigating bias [6 marks]

- during the data collection phase of the ML lifecycle
- during the deployment and monitoring of an ML model.

3 Deep Learning and CNNs [25 marks]

(a) Describe the function of a Convolutional Layer in a Convolutional Neural Network (CNN). [4 marks]

(b) Identify and briefly describe the function of 1,2,3 and 4 labelled in red in Figure 1 below in assuming a standard CNN architecture. [8 marks]

(c) What is the primary purpose of the Backpropagation algorithm when training a Feedforward Neural Network? [5 marks]

(d) Calculate the net input to and ReLU activation for h_1 in ANN depicted in Figure 2 below, assuming the following: [8 marks]

- ReLU: g(z) = max(0,z)
- bias = 0

input $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$ and weights $\begin{bmatrix} 0.5 \\ 1.6 \\ -0.3 \end{bmatrix}$

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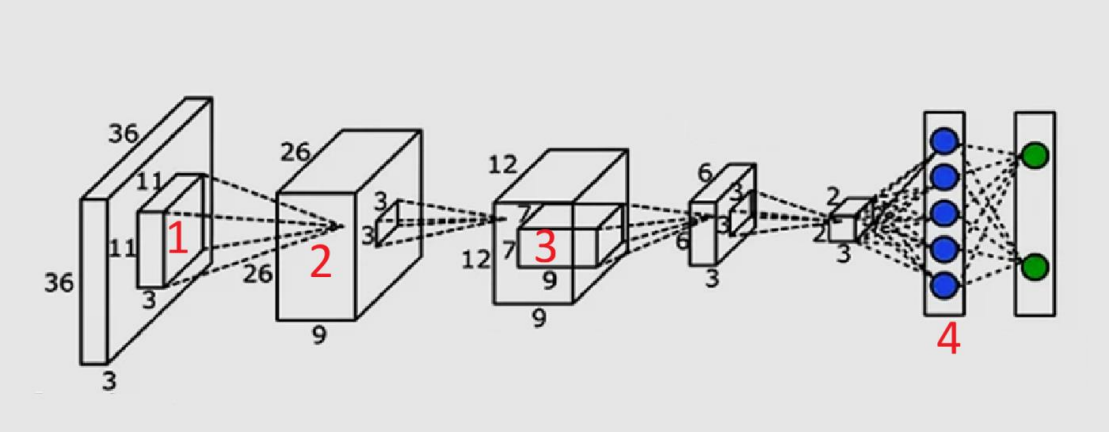


Figure 1 – To be referred to in answers for Question 3b.

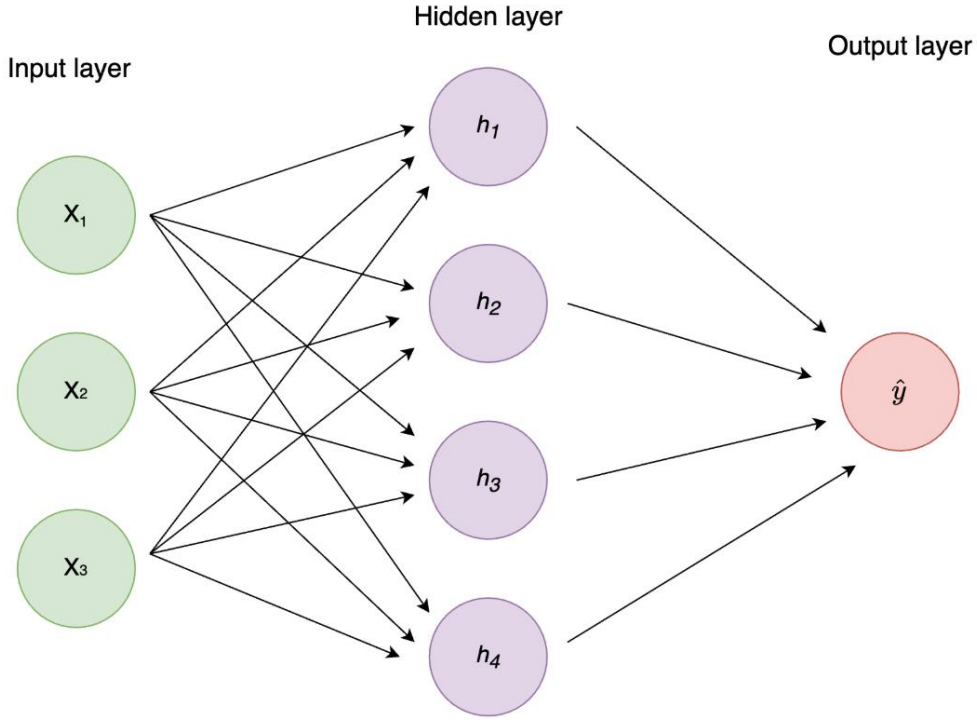


Figure 2 - To be referred to in answers for Question 3d.

4 Data Representation and K-means clustering [25 marks]

(a) Explain using an example why feature scaling might be an important step in multivariable linear regression. [5 marks]

(b) Compare two different feature scaling techniques giving an example of when one might be used over another. [8 marks]

(c) In K-means clustering, why is it not appropriate to choose a value of k that simply minimises Within-Cluster Sum of Squares? [5 marks]

(d) Propose a suitable technique to determine the value of k, noting any relevant limitations. Use a diagram to illustrate your answer. [7 marks]